

Indian sign language to voice using ESP32_Cam for hand tracking and MediaPipe for hand gesture detection

Problem Addressed

Communication between deaf or speech-impaired individuals and people who do not understand sign language remains a major challenge. Many people rely on sign language for communication, but the general population does not know it, which creates communication barriers. Therefore, an automated system is needed to convert sign language gestures into understandable text and speech.

Research Gap

Most existing sign language recognition systems are designed mainly for **American Sign Language (ASL)** and require expensive hardware or complex wearable sensors. Very few systems support **Indian Sign Language (ISL)**, and many of them lack real-time performance and affordability.

Key Contribution

This research proposes a **real-time Indian Sign Language translation system** that uses:

- **ESP32-CAM module** for hand tracking
- **MediaPipe framework** for hand landmark detection
- **Machine learning (Random Forest classifier)** for gesture classification
- **Text-to-Speech (TTS)** for converting recognized text into speech

The system translates ISL gestures into text and then converts the text into voice to enable communication between deaf individuals and others.

Methodology

The system consists of several stages.

▪ Data Acquisition

An external camera captures live video of hand gestures. The **ESP32-CAM module tracks hand movements** and provides the coordinates of the detected hand.

- **Preprocessing**

Video streams are divided into frames, and the frames are converted from **BGR to RGB format** to be processed by the MediaPipe framework.

- **Hand Landmark Detection**

MediaPipe detects **21 hand landmark points** representing the positions of fingers and palm. These landmarks are used as input features for gesture recognition.

- **Gesture Classification**

A **Random Forest classifier** is trained on labeled gesture data to recognize different ISL gestures. The model learns patterns from hand landmark coordinates to classify gestures accurately.

- **Text-to-Speech Conversion**

Once the gesture is recognized, the system converts the corresponding text into **spoken voice using Google Text-to-Speech (TTS)**.

Results

The system was evaluated using performance metrics such as accuracy, precision, recall, and F1-score.

Performance results:

Metric	Value
Accuracy	~95.67%
Precision	~0.98
Recall	~0.96
F1-Score	~0.97

These results indicate that the model can accurately detect ISL gestures and convert them into text and speech in real time.

Advantages of the System

The proposed system provides several benefits:

- **Low-cost hardware implementation** using ESP32-CAM
- **Real-time gesture recognition** using MediaPipe
- **Portable and lightweight design**
- **Scalability**, allowing future expansion to more gestures and languages

These features make the system practical for real-world use by hearing- and speech-impaired individuals.

Conclusion

The proposed system successfully converts **Indian Sign Language gestures into text and speech** using computer vision and machine learning techniques. With an accuracy of around **95.67%**, the system demonstrates the potential to improve communication between deaf individuals and the general public.